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ChromatinHMM: Extending ChromHMM with count-based emission distributions

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Abstract

We present ChromatinHMM, a hidden Markov model framework for chromatin state discovery using three interchangeable emission distributions, applied to GM12878 ChIP-seq data across five histone marks. Holding modeling choices constant except for emission distribution, we find that emission choice leads to distinct trade-offs across evaluation metrics. The Bernoulli model achieves the highest concordance with reference annotations, while the Negative Binomial model provides improved separation of regulatory states and stronger generalization to held-out data. These results show that incorporating count information alone is insufficient to improve chromatin state annotation; gains in interpretability and robustness arise specifically from modeling overdispersion. No single model dominates across all metrics, suggesting that the optimal emission distribution depends on biological context and evaluation criteria. Source code and data available at `https://github.com/tori-chou/ChromatinHMM`.

1. Introduction

Chromatin state annotation partitions the genome into functionally distinct regions based on combinatorial histone mark patterns measured by ChIP-seq. These annotations are widely used to interpret gene regulation, cell-type-specific activity, and disease-associated genetic variants from genome-wide association studies (Roadmap Epigenomics Consortium et al., 2015).

ChromHMM is among the most widely adopted tools for chromatin state discovery (Ernst and Kellis, 2012). It first binarizes ChIP-seq signal at each 200bp genomic bin using a Poisson background model, then fits a multivariate hidden Markov model where each histone mark is modeled as an independent Bernoulli variable. Binarization was introduced for computational tractability and robustness to variation in sequencing depth across experiments (Ernst and Kellis, 2012; Mammana and Chung, 2015), and has enabled large-scale annotation of hundreds of cell types (Roadmap Epigenomics Consortium et al., 2015). However, binarization discards read count magnitude, potentially obscuring quantitative differences between regulatory elements of different strengths.

We present a controlled comparison of Bernoulli, Poisson, and NB emissions within a unified HMM framework, holding all other modeling choices constant. We evaluate the resulting chromatin state annotations on GM12878 lymphoblastoid ChIP-seq data across five histone marks, using biological annotation enrichment, reference segmentation concordance, gene expression correlation, and held-out log-likelihood as evaluation criteria.

2. Methods

We implement a unified ChromatinHMM class with pluggable emission distributions.

2.1. Data acquisition

ChIP-seq alignments for GM12878 were obtained from ENCODE (GRCh38, replicate 1) for five histone marks: H3K4me1, H3K4me3, H3K27me3, H3K36me3, and H3K9me3, with matched input. Gene annotations were from GENCODE v45, CpG islands from UCSC, and Roadmap Epigenomics GM12878 15-state segmentation (E116) was used as reference. Gene expression data were obtained from ENCODE.

2.2. Data preprocessing

The genome was partitioned into 200bp bins. For the Bernoulli model, signals were binarized using ChromHMM v1.27 BinarizeBam (Poisson threshold $p < 10^{-4}$). For Poisson and NB models, raw read counts per bin were extracted using pysam, excluding unmapped and duplicate reads. Count matrices were computed per chromosome. Counts were not explicitly normalized for sequencing depth, as per-mark emission parameters absorb scale differences. Models were trained on chromosomes 17, 19, 21, and 22 ($\sim$940,000 bins).

2.3. Model

ChromatinHMM implements a shared HMM backbone with identical transition initialization, Baum–Welch EM training, and Viterbi decoding across emission types (K = 10 states). Standard scaled forward–backward algorithms were used, with Numba acceleration. Each state models marks independently:

- **Bernoulli:** binary emissions with parameters θ_{km} (closed-form M-step).
- **Poisson:** count emissions with rates λ_{km} (closed-form M-step).

- **NB:** parameterized by mean μ_{km} and dispersion r_{km} (variance $\mu_{km} + \mu_{km}^2/r_{km}$). Means are updated in closed form, while dispersion is estimated numerically via bounded optimization of the weighted log-likelihood.

2.4. Training

Models were trained using batch Baum–Welch EM for up to 2000 iterations, with convergence defined as absolute log-likelihood change < 0.1. To mitigate initialization effects, each model was trained with four random seeds, and the highest-likelihood run was retained.

2.5. Evaluation

Models were evaluated using Viterbi state assignments on training chromosomes and held-out likelihood on chromosome 18.

- **Roadmap ARI:** Agreement with Roadmap E116 segmentation using Adjusted Rand Index. Tentative labels were generated based on overlap with Roadmap states.
- **Mean state purity:** Average maximum Roadmap class proportion per inferred state.
- **TPM dynamic range:** Ratio of maximum to mean median gene expression (TPM) across states.
- **Held-out log-likelihood:** Per-bin likelihood on chromosome 18 for Poisson and NB models.

3. Results and discussion

3.1. Learned emission profiles

Learned emission parameters recapitulate known histone mark patterns and verify tentative labels (Figure 1). Across NB and Poisson models, H3K36me3 correctly marks transcriptional activity, and H3K9me3 marks heterochromatin. The Bernoulli model produces near-binary states with a single active promoter state (H3K4me3/H3K4me1), and single enhancer state (H3K4me1). In contrast, Poisson and NB models capture graded signal differences, but also introduce biologically inconsistent associations in some states. For example, H3K27me3, associated with gene silencing, is strongly enriched in the NB "Enhancer" state, indicating imperfect separation of regulatory signals (Figure 1).

3.2. Concordance with Roadmap reference annotations

Comparison to Roadmap E116 shows that Bernoulli achieves the highest ARI (0.414), followed by Poisson (0.326) and NB (0.228), with Bernoulli and Poisson exceeding ChromHMM (0.293) while NB falls below (Figure 3). However, NB achieves the highest state purity (59.4%), though differences in purity are modest across models. The higher ARI of Bernoulli may be driven by the basis of the Roadmap annotations, which uses ChromHMM's original model of independent Bernoulli variables.

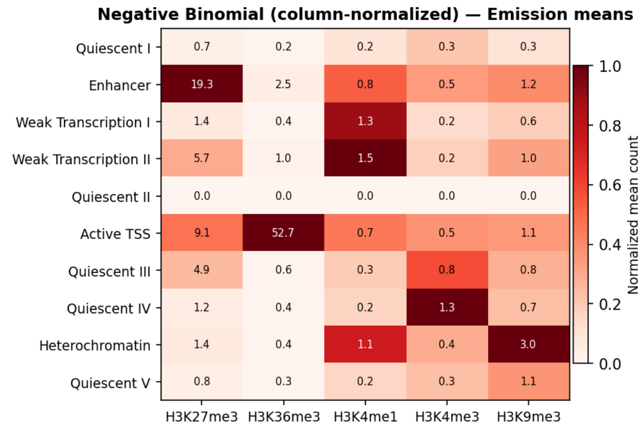

Figure 1 Emission parameter heatmaps for the NB model. Rows are inferred chromatin states labeled with their biological identity; columns are histone marks. Values are column-normalized mean counts. Equivalent figures for Bernoulli and Poisson are provided as supplementary material.

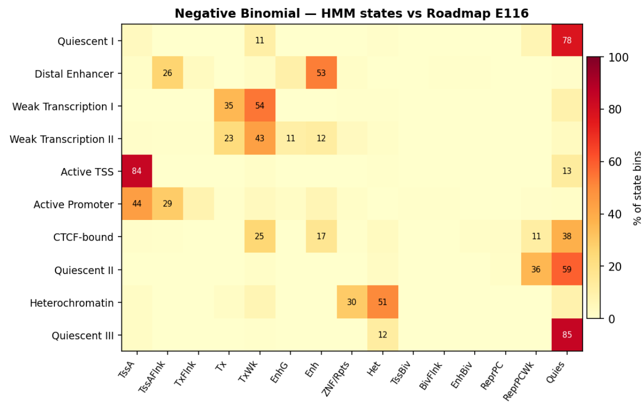

Figure 2 Row-normalized confusion matrix between NB model states and the Roadmap E116 15-state reference annotation. Cell values show the percentage of bins in each model state assigned to each Roadmap class. Equivalent figures for Bernoulli, Poisson, and ChromHMM are provided as supplementary material.

3.3. Enrichment at ENCODE regulatory elements

All models identify and are able to differentiate promoter- and enhancer-associated states based on the Roadmap-inferred labels, and validated with GENCODE annotations, UCSC CpG islands, and Candidate cis-Regulatory Elements by ENCODE (Figure 4). Promoter states show strong enrichment of CpG islands, TSSs, cCRE PLS and cCRE pELS, while enhancer states show enrichment at cCRE dELS. Only the Poisson model produces three promoter-enriched states ($>5\times$), whereas NB, Bernoulli, and ChromHMM each yield a single promoter state. This fragmentation arises from the Poisson mean–variance constraint, which splits bins with similar mark presence but differing counts. The NB model absorbs this variability via dispersion, consolidating promoter signal.

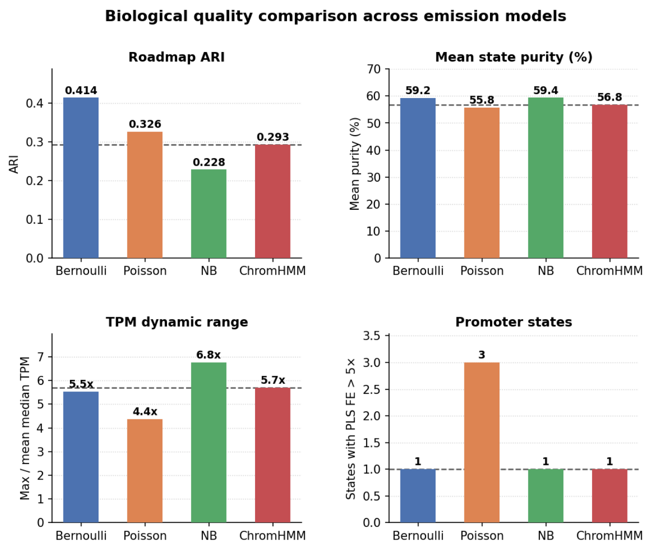

Figure 3 Biological quality metrics across all four emission models. Dashed lines indicate the ChromHMM reference value (panels 1-3). Higher is better for ARI, purity, and dynamic range (expression stratification).

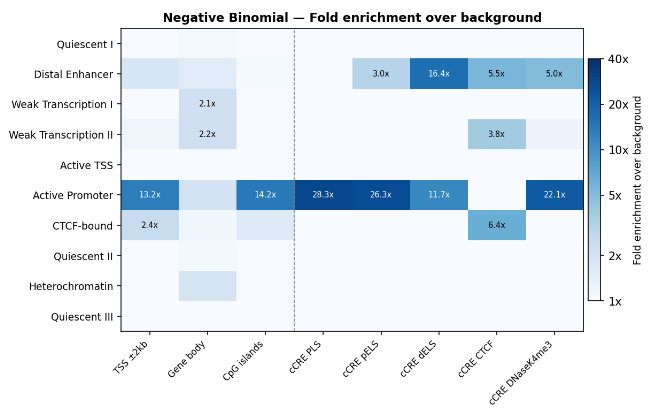

Figure 4 Fold enrichment over genomic background at ENCODE cCRE and positional annotations for the NB model. Colour is $\log_2$ fold enrichment, capped at 40×. The dashed vertical line separates positional annotations (left) from cCRE categories (right). Equivalent figures for all models are provided as supplementary material.

3.4. Gene expression stratification

NB achieves the greatest separation of gene expression across states (6.8×), followed by ChromHMM (5.7×), Bernoulli (5.5×), and Poisson (4.4×) (Figure 5). The reduced range in Poisson is consistent with promoter fragmentation, which distributes highly expressed genes across multiple states. Across models, promoter states show elevated TPM, while quiescent states remain near zero.

3.5. Generalization to held-out data

On chromosome 18, NB achieves a per-bin log-likelihood of -6.41, compared to -689.42 for Poisson. The Poisson model also produces 56 near-zero probability bins, while NB produces none. This reflects Poisson's inability to accommodate

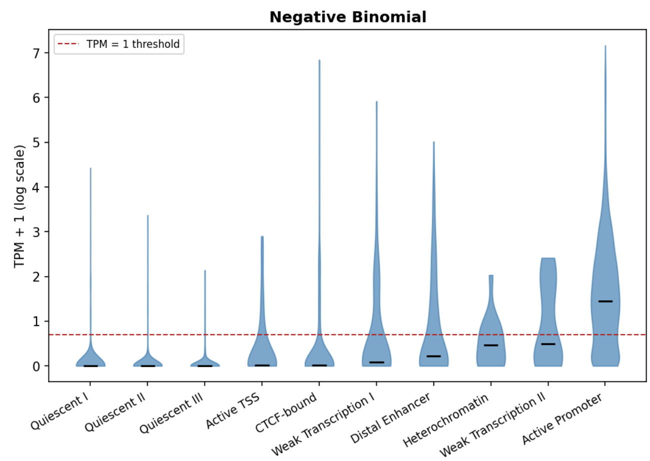

Figure 5 Per-state gene expression distributions for the NB model. States are ordered by median TPM (ascending). The dashed red line marks TPM $= 1$. Equivalent figures for all models are provided as supplementary material.

distribution shifts in count magnitude, whereas NB's dispersion parameter provides robustness to variability. Chromosome 18 exhibits higher H3K4me1 signal and lower H3K27me3 and H3K36me3 relative to the training chromosomes, a shift that the Poisson model cannot accommodate. The NB dispersion parameter provides the necessary flexibility, assigning non-negligible probability to observations that fall outside the range observed during training.

3.6. Overall comparison

The three emission models exhibit distinct trade-offs across evaluation metrics. Bernoulli achieves the highest concordance with Roadmap annotations, likely reflecting the shared modeling assumptions between ChromHMM-based annotations and independent Bernoulli emissions. Bernoulli produces biologically coherent states and remains competitive across most metrics.

The NB model provides the strongest separation of regulatory states, achieving the greatest separation of gene expression across states and highest state purity. Its ability to model overdispersion allows it to capture variability in read counts and produce more graded distinctions between chromatin states. NB also demonstrates substantially improved generalization to held-out data, indicating robustness to distribution shifts in count magnitude.

In contrast, the Poisson model underperforms across several dimensions. While it achieves moderate agreement with reference annotations, it fragments promoter-associated states and generalizes poorly to held-out chromosomes. These limitations arise from the Poisson assumption of equal mean and variance, which prevents it from accommodating overdispersed ChIP-seq data.

4. Availability and implementation

ChromatinHMM is implemented in Python and supports Bernoulli, Poisson, and NB emission distributions within a unified hidden Markov model framework. The software is available at https://github.com/tori-chou/ChromatinHMM, along with documentation and example workflows for reproducing all experiments in this study. Supplementary figures are also provided in the GitHub repository.

The implementation relies on standard scientific Python libraries including NumPy, SciPy, and pysam, and is compatible with Linux and macOS systems. No specialized hardware is required.

5. Conclusion

We implemented Bernoulli, Poisson, and NB emission distributions within a unified HMM framework for chromatin state discovery and evaluated them on GM12878 ChIP-seq data. We find that emission choice leads to distinct trade-offs across evaluation metrics. The Bernoulli model achieves the highest concordance with reference annotations and remains competitive despite discarding count magnitude. In contrast, the NB model provides improved separation of regulatory states and stronger generalization to held-out data, reflecting its ability to model overdispersed count distributions. The Poisson model underperforms across several dimensions, as its mean–variance constraint leads to state fragmentation and poor robustness to distribution shift. These results demonstrate that incorporating count information alone is insufficient to improve chromatin state annotation. Instead, gains in interpretability and generalization arise specifically from modeling overdispersion. Overall, the choice of emission distribution has measurable consequences for chromatin state quality, and careful modeling of count variability is important for producing biologically meaningful annotations.

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